

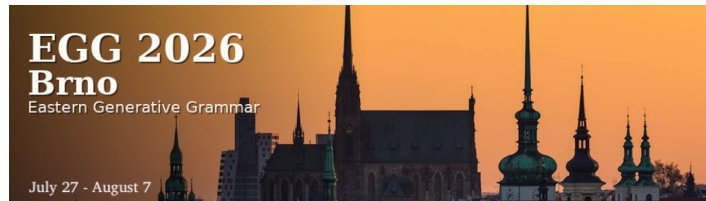


Minimalist Foundations

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<https://kxhoge.github.io/resources.html>



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About this course

- *What does this course address?*

It has been over thirty years since the publication of 'A Minimalist program for linguistic theory' (Chomsky 1993; reprinted in 1995) inaugurated a programme of syntactic theorising that is guided by the assumption that the architecture of natural language should be reduced to the barest essentials. Minimalism, which emerged from Government and Binding theory (Chomsky 1981), is now the longest-running theoretical framework within the Chomskyan tradition of generative syntax. Despite its apparent continuity, Minimalism has undergone substantial internal shifts. This course charts these shifts by tracing how successive phases of Minimalist theorising have progressively redefined the role of Merge, from one syntactic operation among others to a candidate for the sole language-specific computational mechanism.

About this course (cont.)

- *What will I have learnt by the end of this week?*

Upon completion of the course, you will have

- gained an understanding of the historical development of Minimalism from its origins in Government and Binding theory to contemporary approaches and of the major conceptual shifts that have shaped the programme;
- honed your ability to critically evaluate Minimalist arguments, assumptions, and theoretical proposals;
- acquired sufficient familiarity with the theoretical background to be able to understand and situate syntactic research conducted within the Minimalist Program.

About this course (cont.)

- *How is this course taught?*

The course is taught in five lectures. You are welcome to ask clarification and follow-up questions during the lectures. Since syntax is best learnt through hands-on engagement, each session will feature a brief practical segment, which you can tackle either independently or by collaborating with your seat neighbour.

Course outline

Monday	From X-bar theory to Bare Phrase Structure: The emergence of Merge
Tuesday	External and Internal Merge
Wednesday	Constraining Merge: Phase theory
Thursday	The Strong Minimalist Thesis, Merge, and the problem of labelling
Friday	The Miracle Creed: Can syntax be reduced to Merge?

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Outline

1. Learnability, universality, and evolvability
2. Simplify, simplify, simplify!
3. Minimalist structure building
4. Summary and outlook

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- Two central observations have stood at the heart of the generative enterprise since its inception:
 - **linguistic creativity**: the capacity of speakers to produce and interpret an unbounded number of (hierarchically structured) linguistic expressions;
 - **‘linguistic promiscuity’**: ‘the fact that any child (absent pathology) acquires any language in (roughly) the same way as another child does’ (Hornstein 2018: 50).
- Together, linguistic creativity and linguistic promiscuity suggest that the human capacity for language is a property of our biological endowment; that is, we possess

‘a computational system coded in the brain that for each individual recursively generates an infinite array of hierarchically structured expressions, each formulating a thought, each potentially externalized in some sensory-motor (SM) medium – what we may call the *Basic Property of Language*’ (Chomsky 2021: 7).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- Seeking to develop an explanatory theory of language, the task is twofold:
 - i. to explain the properties of individual I-languages—that is, speakers' language faculties in their mature state—and the systems by which they are externalised;
 - ii. to determine the initial state of the language faculty (**Universal Grammar**), namely the innate properties that make the acquisition of any human language possible.

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- ‘UG has goals that appear contradictory. It must meet at least three conditions:
 - (i) It must be rich enough to overcome the problem of *poverty of stimulus* (POS), the fact that what is acquired demonstrably lies far beyond the evidence available.
 - (ii) It must be simple enough to have evolved under the conditions of human evolution.
 - (iii) It must be the same for all possible languages, given commonality of UG.’
(Chomsky 2021: 7)

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- Learnability and **evolvability** pose competing requirements: while the POS provides an argument for a UG with considerable complexity, the relatively short evolutionary history of language (standardly assumed to encompass no more than 100,000 years) suggests that UG must be sufficiently simple to have evolved in the required time frame.

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- ‘The evidence for [linguistic recency] is twofold ...

First, ... if one assumes that cultural artifacts are proxies for the emergence of language, then evidence of rich cultural artifacts is very recent, roughly 50-100 kya ... A common conclusion is that the efflorescence of such artifacts is tied to the emergence of language in humans.

[Second], FL arose in humans before the trek out of Africa [ca. 70,000-60,000 years ago] and has remained unchanged since. ... As far as we know, there is no robust genetic, ethnic, racial or areal variation in the acquisition of any language.

... Both facts point to the same conclusion: that whatever prompted the emergence of FL is recent and very minor ... To put this yet another way: The less UG-ish FL is, the easier it is to explain how it arose in the species so recently (fact 1) and why it has remained largely unchanged (fact 2).’ (Hornstein 2018: 53).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- FLs with simpler UGs—that is, with less language-specific machinery—may also be **methodologically** preferable, on the assumption that much of cognitive architecture is shared across domains.

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Learnability, universality, and evolvability

- ‘In summary, both methodological and evolutionary reasoning support a project wherein the UG features of FL are minimized, hence the MP program: Determine to what degree a more computationally generic domain-general view of FL is viable’ (Hornstein 2018: 54).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Simplify, simplify, simplify!

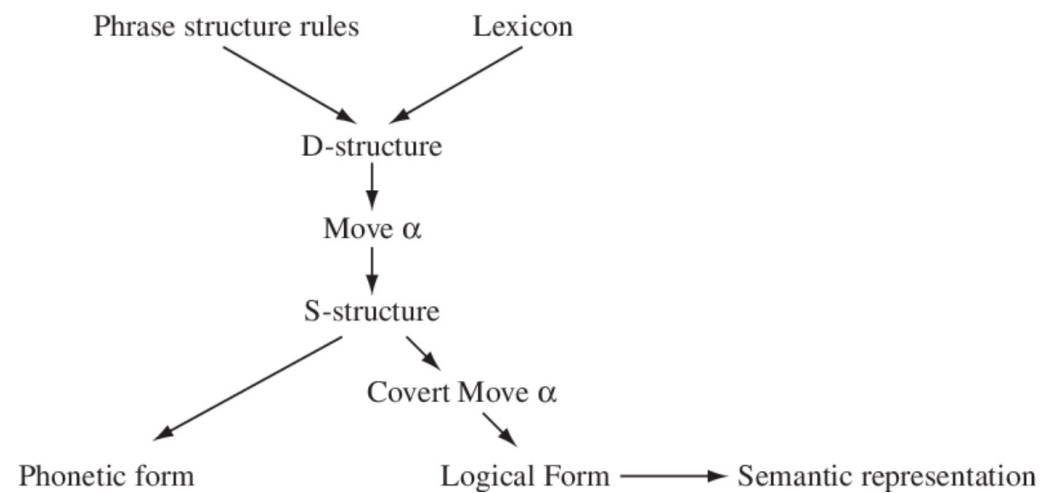
- ‘Practically speaking, how does one pursue this MP agenda? The answer is that one should start with GB and simplify, simplify, simplify’ (Hornstein 2018: 54).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Simplify, simplify, simplify!

- Architecture of Government and Binding theory

(1)



(Jackendoff 2010: 68)

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Simplify, simplify, simplify!

- Government and Binding theory further conceives of grammar as consisting of interacting modules ('subsystems of principles'), each governed by its own set of principles.

- (2)
- (i) bounding theory
 - (ii) government theory
 - (iii) θ -theory
 - (iv) binding theory
 - (v) Case theory
 - (vi) control theory

(Chomsky 1981: 5)

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Simplify, simplify, simplify!

- Simplifying the architecture of grammar:

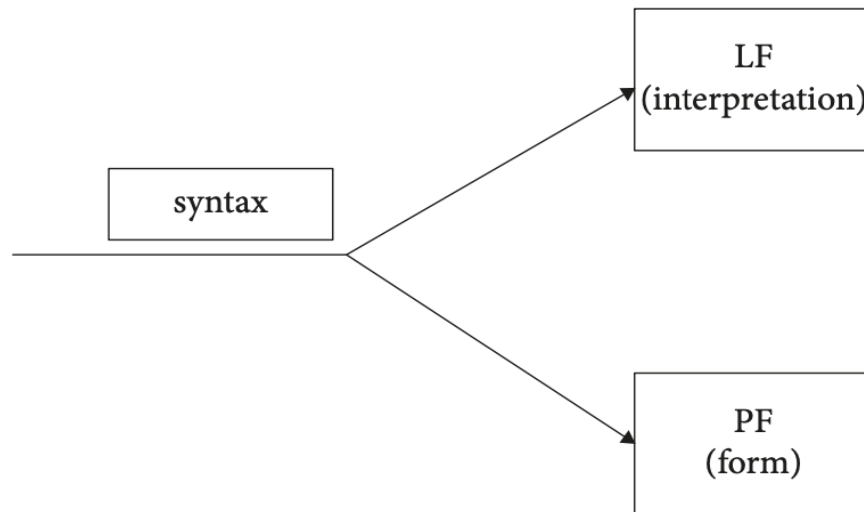
‘UG must determine the class of possible languages. It must specify the properties of S[tructural]D[escription]s and of the symbolic representations that enter into them. In particular, it must specify the interface levels (A[rticulatory]-P[erceptual], C[onceptual]-I[n]tentional), the elements that constitute these levels, and the computations by which they are constructed. **A particularly simple design for language would take the (conceptually necessary) interface levels to be the only levels.** That assumption will be part of the “minimalist” program I would like to explore here’ (Chomsky 1995: 155; emphasis added).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Simplify, simplify, simplify!

- The Minimalist Y-model of grammar

(3)



(Wiltschko & Heim 2016: 330, Fig. 17))

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Simplify, simplify, simplify!

- Ontological parsimony also extends to the subsystems of principles, raising the question of whether the independent modules of GB are necessary or whether their effects can be derived from a more economical computational system.

‘... the first step is to unify the various modules ... and show that they are all instances of the same underlying operations and principles. This wing of the MP project has antecedents in earlier GB research; the prime example is the unification of Ross’s island effects in terms of Subjacency Theory ...

Second, we show that the principles and operations of the unified theory (the results of step 1) have (very) few linguistically specific operations and principles. In other words, we show that the required operations and principles that the unification invokes are large computationally / cognitively generic’ (Hornstein 2018: 54).

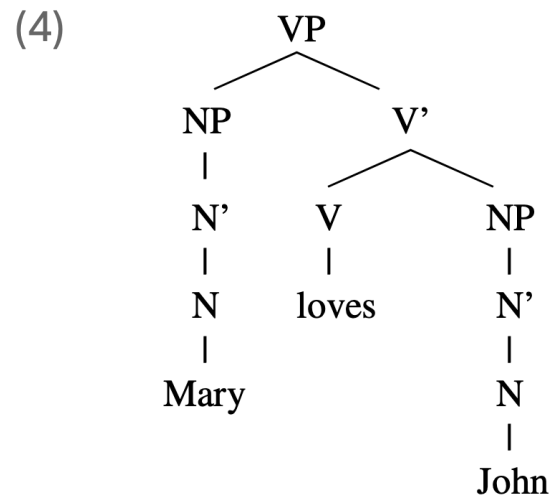
From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- The Minimalist commitment to ontological parsimony motivates a re-examination of X-bar theory.
- Chomsky (1995: 228) proposes in its place **bare phrase structure** theory, in which ‘categories are elementary constructions from properties of lexical items; ... there are no bar levels and no distinction between lexical items and “heads” projected from them’.

From X-bar theory to Bare Phrase Structure: The emergence of Merge

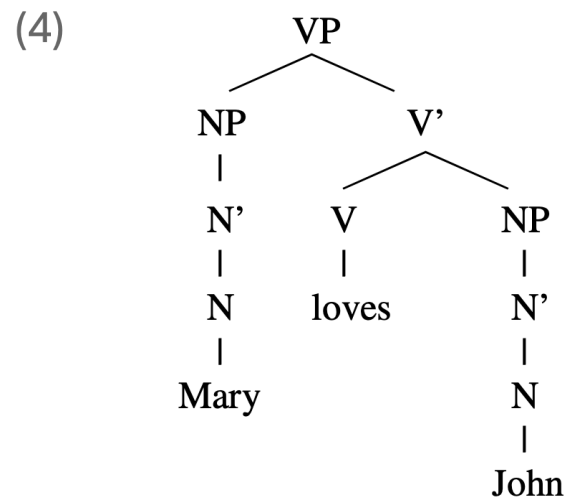
Minimalist structure building



- **Problem 1:** The vacuous projection of lexical items to X' introduces a representational object (the bar level) that is not a property of the lexical items themselves (and therefore incompatible with the Inclusiveness Condition).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building



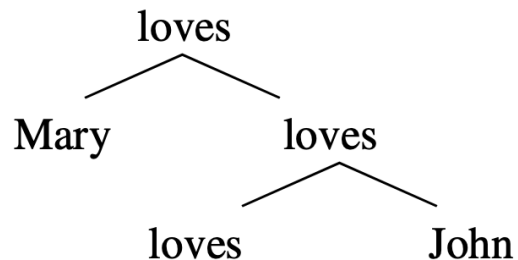
- **Problem 2:** Lexical items already encode their categorial properties, rendering the preterminal category nodes that dominate them redundant.

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- In bare phrase structure, the status of a syntactic object as minimal, intermediate, or maximal is determined by its structural relations within the syntactic object:
 - minimal projections are lexical categories that do not dominate any other syntactic object;
 - maximal projections are syntactic objects that do not project further;
 - intermediate projections are syntactic objects that are neither minimal nor maximal.

(5)



From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- *Try your hand at some syntactic analysis*

- Give the bare phrase structure representation for

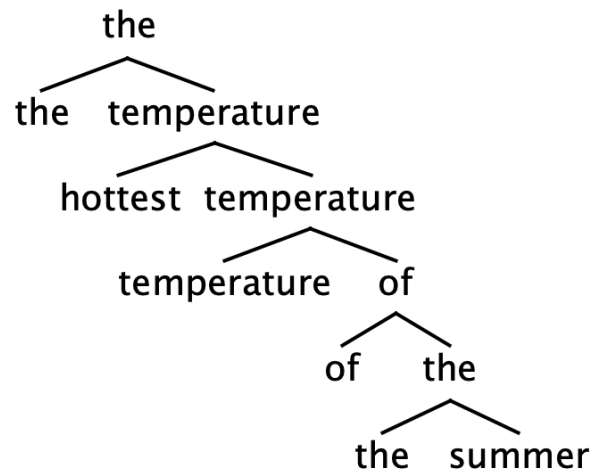
the hottest temperature of the summer.

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- *Try your hand at some syntactic analysis*

(6)



From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- Under early Minimalist assumptions, a syntactic derivation is generated from a **numeration** by the **selection** of lexical items and the structure-building operation **Merge**.
 - ‘Let us take a **numeration** to be a set of pairs (LI, i) , where LI is an item of the lexicon and i is its index, understood to be the number of times that LI is selected’ ...
 - Given the numeration N , the operations of C_{HL} recursively construct syntactic objects from items in N and syntactic objects already formed ...
 - One of the operations of C_{HL} is a procedure that selects a lexical item LI from the numeration, reducing its index by 1, and introduces it into the derivation as SO_{n+1} . Call the operation **Select** ...
 - At the LF interface, Σ [the output generated by a syntactic derivation] can be interpreted only if it consists of a single syntactic object. Clearly, then, C_{HL} must include a second procedure that combines syntactic objects already formed ...
 - The simplest such operation takes a pair of syntactic objects (SO_i, SO_j) and replaces them by a new combined syntactic object SO_{ij} . Call this operation **Merge**’
(Chomsky 1995: 207-208).

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Minimalist structure building

- **Merge** is a binary operation that forms a new syntactic object by combining two syntactic objects into an unordered set; the resulting object receives a label through the projection of one of the merged elements.

‘Applied to two objects α and β , Merge forms the new object K , eliminating α and β ... The simplest object constructed from α and β is the set $\{\alpha, \beta\}$, so we take K to involve at least this set, where α and β are the constituents of K . Does that suffice? Output conditions dictate otherwise; thus, verbal and nominal elements are interpreted differently at LF and behave differently in the phonological component. K must therefore at least (and we assume at most) be of the form $\{\gamma, \{\alpha, \beta\}\}$, where γ identifies the type to which K belongs, indicating its relevant properties. Call γ the label of K ’ (Chomsky 1995: 223).

‘...the label γ is either α or β ; one or the other projects and is the head of K . If α projects, then $K = \{\alpha, \{\alpha, \beta\}\}$ ’ (Chomsky 1995: 224).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- Not all structure building involves Merge: in early Minimalism, **Move** remains a distinct operation responsible for creating displacement dependencies.

‘While Merge is costless for principled reasons, movement is not: the operation takes place only when forced (Last Resort); and it is overt, violating Procrastinate, only when that is required for convergence. If α has a strong feature F, it triggers an operation OP that checks F before the formation of a larger category that is not a projection of α . The operation OP may be Merge or Move’ (Chomsky 1995: 216).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Minimalist structure building

- Move takes place only when forced (Last Resort).
- If α has a strong feature F, it triggers an operation (Merge or Move) that checks F before the formation of a larger category that is not a projection of α .

- (7) a. there is [α a strange man] in the garden
b. there seems to [α a strange man] [that it is raining outside]
(Chomsky 1995: 183, ex. (26))

- In both (7a) and (7b), the strong D-feature on I is satisfied by Merge of *there*.
- In (7a), *a strange man* is not in a proper position for Case checking and so is forced to move to SpecIP at LF, adjoining to *there*.
- In (7b), *a strange man* checks Case inside the PP and so cannot raise to SpecIP, with the result that freestanding *there* receives no semantic interpretation.
'The derivation thus converges, as semi-gibberish' (Chomsky 1995: 184).

From X-bar theory to Bare Phrase Structure: The emergence of Merge

Summary and outlook

- Guided by the problem of language evolution ('Darwin's problem'), the Minimalist Program pursues an increasingly parsimonious theory of language, seeking to eliminate representational and computational complexity in the search for the simplest possible account of the human language faculty.
- In its initial formulation (Chomsky 1995), the Minimalist Program introduces Merge as the fundamental operation for constructing hierarchical syntactic structure, retaining Move as a distinct operation responsible for deriving displacement.
- Evidence for displacement is widespread in natural language. What challenge does the fact that phrases are interpreted both where they are heard and in positions that are not articulated pose for the Minimalist agenda?

From X-bar theory to Bare Phrase Structure: The emergence of Merge

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